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Faculty Of Pharmacy

Phase One/ Semester 2

Main points of the last lecture

In the last lecture, we discuss:

- **H**istory of Pharmacognosy,
- Importance and uses of plants in medicine & pharmacy,
- Types of drugs derived from plants,

- **C**lassification of vegetable drugs,
- **O**rigins of herbal drugs,
- **B**inomial system for naming the plants.

Starch

Outlines

1. Introduction
2. Chemical entity of the starch
3. Physical properties of the starch
4. Breakdown of CHO
5. Uses of starch
6. Starch based products

Objectives of the lecture

By the end of this lecture, the student should be able to:

1. **D**efine starch,
2. **R**ecognize chemical entity of the starch,
3. **R**ecognize physical properties of the starch,
4. **E**xplain breakdown of CHO,

5. **M**ention uses of starch in medicine,

6. **R**ecognize starch based products and their properties.

1. **Introduction**

2. Chemical entity of the starch

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Introduction

Starch is a homopolysaccharide used by some plants as energy store; there are 2 components of starch:

1. Amylose: chains of glucose residue connected by α-(1-4) glycosidic bonds, amylose chains While amylose was traditionally thought to be completely unbranched, it is now known that some of its molecules contain a few branch points

2. Amylopectin : similar to amylose (it contains glucose residues linked by α-(1-4) glycosidic bond) BUT in amylopectin this chain branched through additional α-(1-6) glycosidic bond.

Amylose is a much smaller molecule than amylopectin.

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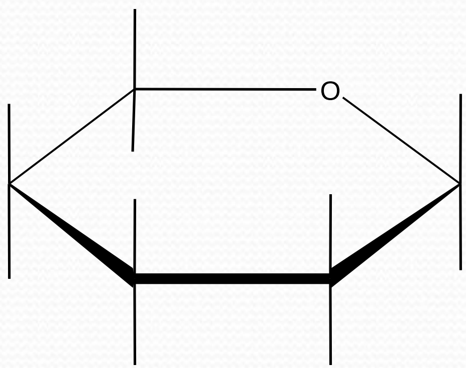
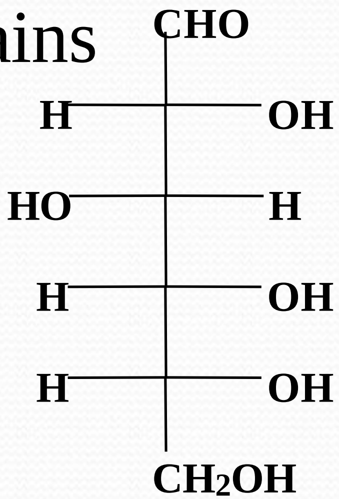
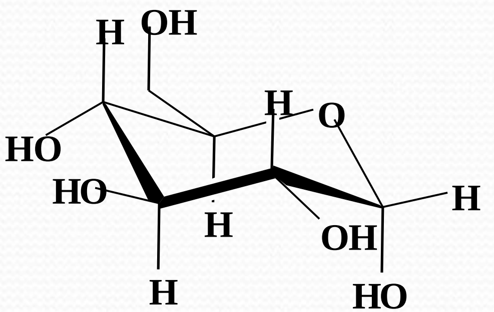
Starch chemically

Ring structures



Open chains





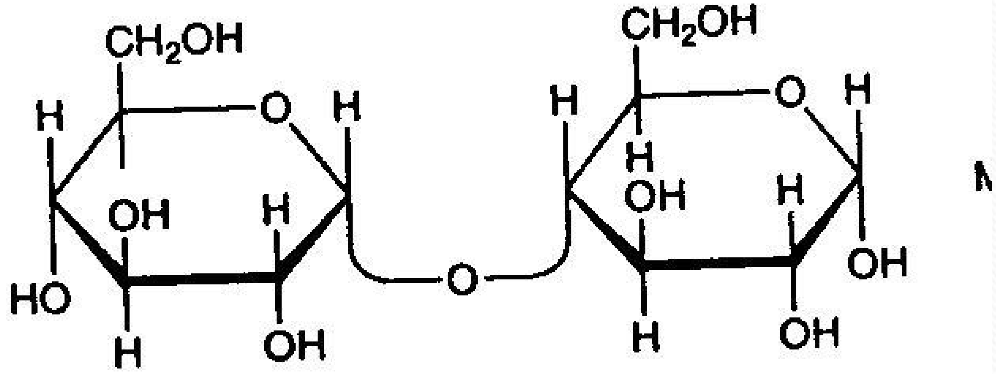
chair

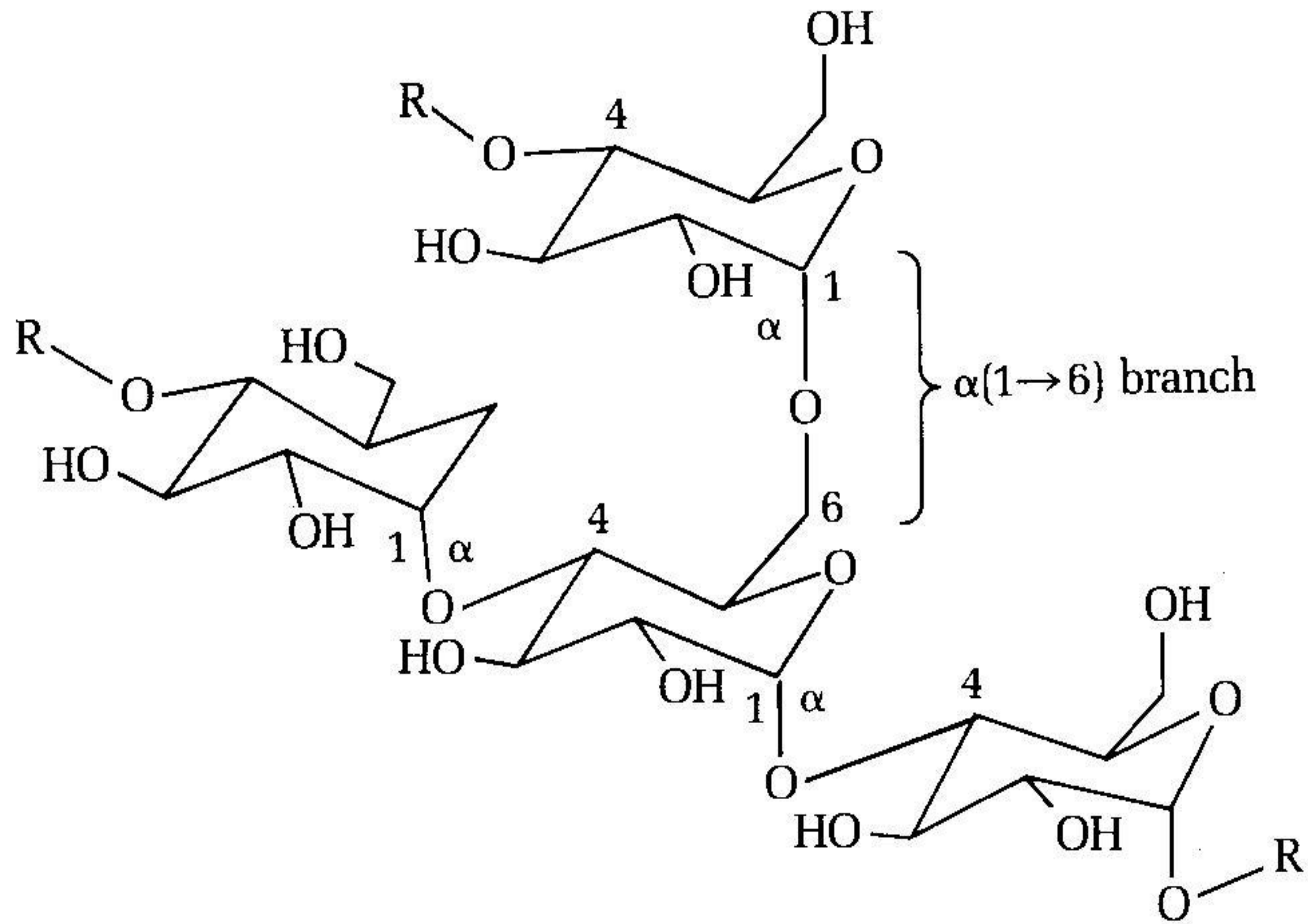
Starch the chemical

Starch is glucose polymers ((Amylose (α 1-4) & Amylopectin (α 1-4 & α 1-6))).

It is packed into granules different size and shape depending of the botanical source semi crystalline structures

Link α (1-4) :





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Link α (1-6) :

	Wheat	Maize	Potato
• Amylose	26-31	24-32	23

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Physical properties

- Starch is biosynthesised in an aqueous environment
- Drying starch causes shrinkage and cracks.
- Dry starch in water can absorb up to 50% of its weight & expand 30 -100% (reversible).

- Starch molecules arranged in the plant in semi-crystalline granules. Each plant species has a unique starch granular size: rice starch is relatively small (about 2μm) while potato starches have larger granules (up to 100μm).

- Starch becomes soluble in water when heated. The granules swell and burst, the semi-crystalline structure is lost and the smaller amylose molecules start leaching out of the granule, forming a network that holds water and increasing the mixture's viscosity. This is called starch gelatinization

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Breakdown of CHO

The enzymes that break down or hydrolyze starch into the constituent sugars are known as amylases. α-amylases are found in plants and in animals. Human saliva is rich in amylase, and the pancreas also secretes the enzyme. β-amylase cuts starch into maltose units.

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Uses of starch in medicines

- Major food for animals
- Used in paper industry
- Filler component for many pharmaceuticals
- Some times eaten by people

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Starch based products

- Potatoes
- Maize
- Rice
- Wheat
- Cassava

Maize starch

- Maize starch prepared from maize.
- It has chemical formula $(C_6H_{10}O_5)n$.
- It is also know as corn starch or flour.
- Its granule are simple with cleft at center.
- It is more or less neutral.

Uses:

1. Maize starch is most commonly used as a thickening and gelling agent.
2. A fundamental ingredient in most of the packaged food and industrial products

Rice starch

- It is a fine-grained, tasteless white powder obtained by processing raw rice.
- It is inexpensive, hypoallergenic and non-toxic.

Uses

1. Rice starch is an important ingredient in body care products because it has no aroma.
2. Other products made with rice starch include medicines, baby food.

Potato starch

- It is starch extracted from potatoes. The cells of the root tubers of plant contain starch grains (leucoplasts).

Properties

Potato starch contains typical large oval spherical granules, their size ranges between 5 and 100 μm. It is very refined starch, contains minimal protein or fat that gives the powder a clear white colour. Cooked starch has typical characteristics of neutral taste, good clarity.

Extraction

- To extract the starch, the potatoes are crushed; the starch grains are released from the destroyed cells. The starch is then washed out and dried to powder.

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References

- Trease and Evans Pharmacognosy. By W.C. Evans. Published by W.B. Saunders 16th, edition Edinburgh London 2009.
- Textbook Of Pharmacognosy. By T. E. Wallis. Published by S. K. Jain Publishers and distributorsPvt. Ltd., for CBS 5th, edition, London 2005.
- Web sites

Thankyoooo

For

your

listening